

CCNA

01

CCNA — Routing and Switching

Build the routing and switching foundation that every network engineer stands on.

DURATION	LEVEL	MODULE	UPDATED
11 Sessions	Beginner to Intermediate	Module 1	July 2026

IN THIS MODULE

01	Static Routing Concepts
02	Introduction to Dynamic Routing
03	Configuring RIP
04	Introduction to OSPF
05	Configuring OSPF
06	Wireless Standards & Configuration
07	Switching Basics
08	VLAN Concepts
09	VLAN Trunking and Access Ports
10	Inter-VLAN Routing Basics
11	Weekly Review and Practice

SESSION

01

90 MIN · ROUTING FUNDAMENTALS

Static Routing Concepts

◆ Learning Objectives

- Explain what a routing table is and why routers need one
- Define static routing and describe how it differs from directly connected routes
- List the advantages and disadvantages of static routing
- Configure a static route between two routers in Cisco Packet Tracer
- Verify a static route using show ip route and troubleshoot a missing route
- Identify real-world scenarios where static routing is still the right choice

01 Why Routers Need a Map

A router's only job is to move a packet from one network to another. To do that, it checks a routing table — a list of destination networks and the path to reach each one. Without an entry for a network, the router simply drops the packet. Think of the way a BRTS bus driver in Ahmedabad follows a fixed route sheet: no route sheet, no way to know which stop comes next.

◆ Directly Connected vs Remote Networks

A router automatically knows about networks attached to its own interfaces — these show up in the routing table the moment an interface is enabled with an IP address. But for any network beyond that first hop, the router needs help. That help comes either from a human typing in a static route, or from a routing protocol learning it dynamically. Session 01 is about the manual option.

CONCEPT — What Is a Static Route

A static route is a path to a remote network that an administrator types in by hand, rather than one the router discovers on its own. For example, a TOPS lab network with two routers — R1 in the CG Road server room and R2 in a branch office — needs a static route on each side so traffic knows how to cross between them.

Aspect	Static Routing	Where It Matters
Setup effort	Manual, one command per route	Small, stable networks
CPU/Memory use	Very low — no calculations	Low-end routers, WAN links
Adapts to change	No — admin must update it	Networks that rarely change
Security	No route advertisements to intercept	Sensitive point-to-point links

PRO TIP

Static routes always win over dynamic ones learned for the same destination, because they have a lower administrative distance (1, compared to 90–170 for most dynamic protocols). If a route seems 'stuck', check for a leftover static entry first.

Static Routing

Manually configured, fixed path, no extra bandwidth used for updates, but doesn't adjust if a link goes down.

Dynamic Routing

Learned automatically via a routing protocol, adjusts to link failures, but uses CPU and bandwidth to exchange updates.

TRY THIS LIVE

In Packet Tracer, build two routers (R1, R2) connected by a serial or Ethernet link, each with one LAN attached. Configure a static route on R1 pointing to R2's LAN, and vice versa. Ping across both LANs to confirm reachability.

```
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Adds a static route to the 192.168.2.0/24 network via next-hop 10.0.0.2

```
Router(config)# ip route 192.168.2.0 255.255.255.0 Serial0/0/0
```

Alternative form: route out a specific exit interface instead of a next-hop IP

```
Router# show ip route static
```

Displays only the statically configured routes in the routing table

```
Router# show ip route
```

Displays the full routing table, showing the 'S' code next to static entries

02 Static Routing in the Real World

CASE STUDY — Bank Branch Networks in India

Context — Regional and cooperative banks across Gujarat often connect a small branch office to a regional data centre over a single leased line or MPLS link.

Challenge — With only one path in and one path out, running a full dynamic routing protocol added unnecessary CPU load and complexity on low-end branch routers.

What they did — Network teams configured a static default route on the branch router pointing to the regional data centre, and a static route back to the branch subnet on the data centre-side router.

Result — The branch stayed reachable with minimal router overhead, and any change to the link only required a one-line update instead of protocol reconfiguration.

Takeaway — When there is exactly one path to a destination, static routing is often simpler and more reliable than dynamic routing.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — Static Routing chapter.

TYPICAL ADMINISTRATIVE DISTANCE OF A STATIC ROUTE

1

Source: Cisco IOS documentation — administrative distance defaults table.

“A static route is a promise you make to the router — and a promise you have to keep updating yourself.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Kharagpur — Prof. Ajit Pal — Routing fundamentals module

Link — <https://nptel.ac.in/courses/106105080>

Always link the official channel — beware fan-made or re-upload copies.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a friendly CCNA trainer. Explain the difference between a static route and a default route in simple Hinglish, using a Gujarat branch-office-to-head-office network as the example. Keep it under 150 words.

WHERE TO GO NEXT

You now know how to manually tell a router where to send traffic. Next, we look at what happens when the network is too large to configure by hand.

Explore deeper — Read Cisco's 'Static and Default Route' guide on the Cisco Networking Academy site.

Try on your own — In Packet Tracer, remove one static route and observe exactly which pings stop working — this builds your troubleshooting instinct.

Connect the dots — Session 02 introduces dynamic routing, which solves the scaling problem of static routes.

Going further — Research 'floating static routes' — a backup static route with a higher administrative distance, used for automatic failover.

SESSION

02

90 MIN · ROUTING FUNDAMENTALS

Introduction to Dynamic Routing

◆ Learning Objectives

- Define dynamic routing and explain how routers exchange route information
- Compare static routing with dynamic routing across cost, effort, and scalability
- Describe the difference between distance-vector and link-state protocols at a high level
- Observe RIP routing updates in a pre-configured network
- Explain what convergence means and why faster convergence matters
- Identify which real networks benefit most from dynamic routing

01 When Typing Routes By Hand Doesn't Scale

Imagine a college network with 40 routers connecting labs, hostels, and admin blocks across a campus. Configuring static routes for every possible path between every router would take days — and one link failure would mean re-typing routes across the network. This is exactly the problem dynamic routing was built to solve.

CONCEPT — What Is Dynamic Routing

Dynamic routing lets routers talk to each other using a routing protocol, automatically sharing which networks they can reach. If a link goes down — say, a fibre cut near SG Highway — nearby routers detect the change and recalculate a new path, without any admin typing a single command.

◆ Two Broad Families of Protocols

Distance-vector protocols like RIP judge a path mainly by hop count — the number of routers a packet must cross. Link-state protocols like OSPF build a full map of the network and calculate the best path based on link cost. RIP is simple to configure, OSPF is smarter but needs more setup — a trade-off you'll feel across the next three sessions.

Static Routing

Manual, predictable, low overhead, but breaks silently when a link fails and no one updates it.

Dynamic Routing

Self-healing and scalable, automatically adapts to failures, but uses CPU, bandwidth, and needs protocol expertise to tune.

Protocol	Type	Metric Used
RIP	Distance-vector	Hop count (max 15)

Protocol	Type	Metric Used
OSPF	Link-state	Cost, based on bandwidth
EIGRP	Advanced distance-vector	Bandwidth, delay, load, reliability

TRY THIS LIVE

Open the pre-configured RIP network in Packet Tracer. Use `show ip route` on each router and identify which routes are marked 'R' for RIP-learned, versus 'C' for directly connected.

<code>Router# show ip protocols</code>	Shows which routing protocol is running and its key timers
<code>Router# show ip route</code>	Displays the routing table — look for the 'R' code marking RIP routes
<code>Router# debug ip rip</code>	Shows live RIP updates being sent and received (use only in a lab, never production)

02 Convergence — Why Speed Matters

Convergence is the time it takes for every router in a network to agree on the current best paths after something changes. A slow-converging network means dropped calls, stalled payments, or frozen video during that gap. This is one of the biggest reasons networks graduated from RIP to faster protocols like OSPF over the years.

CASE STUDY — ISP Backbone Redundancy in India

Context — Indian ISPs run large backbone networks connecting cities like Ahmedabad, Surat, and Vadodara, often with multiple redundant fibre paths.

Challenge — A single fibre cut — common during road digging or monsoon damage — should not take down connectivity for an entire region.

What they did — ISPs deploy dynamic routing protocols across backbone routers so that when one path fails, traffic reroutes automatically through the remaining fibre paths within seconds.

Result — Customers typically experience a brief hiccup rather than a full outage, because the network converges on a backup path faster than a human could respond.

Takeaway — Dynamic routing turns network failures from 'call the vendor and wait' into 'the network fixes itself'.

Source: Cisco Networking Academy, CCNA — Dynamic Routing Concepts module.

MAXIMUM HOP COUNT SUPPORTED BY RIP

15 hops

Source: RFC 2453 — RIP Version 2 specification.

“Static routes remember what you told them. Dynamic routes remember what the network just taught them.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks and Internet Protocol, IIT Kharagpur — Prof. Soumya Kanti Ghosh & Prof. Sandip Chakraborty — Routing protocols module

Link — <https://nptel.ac.in/courses/106105183>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a patient networking mentor. Explain distance-vector vs link-state routing in simple Hinglish using the analogy of asking a neighbour for directions vs reading a full city map. Keep it under 150 words.

WHERE TO GO NEXT

You've seen dynamic routing at a concept level. Next, you'll configure your first dynamic routing protocol — RIP — hands-on.

Explore deeper — Look up 'RIP vs RIPv2' to understand why classless routing support was added.

Try on your own — Predict, before Session 03, what you think will happen if you disconnect one router's link in a RIP network — then test it.

Connect the dots — Session 03 walks through full RIP configuration on a multi-router network.

Going further — Research why RIP is now considered mostly a legacy/teaching protocol, and where it has been replaced by OSPF or EIGRP in production networks.

SESSION

03

90 MIN · ROUTING FUNDAMENTALS

Configuring RIP

◆ Learning Objectives

- Explain RIP concepts: hop count, periodic updates, and convergence
- Configure RIP on a multi-router network in Packet Tracer
- Enable RIPv2 and understand why it replaced RIPv1
- Capture and analyze RIP updates using simulation tools
- Identify common RIP configuration mistakes and how to fix them
- Explain the limitations that make RIP unsuitable for large networks

01 RIP — The First Routing Protocol Most Engineers Learn

RIP (Routing Information Protocol) is the simplest dynamic routing protocol still taught today. It picks the best path purely by counting hops — the number of routers a packet must pass through. A path through 2 routers always beats a path through 4, even if the 4-router path has faster links. This simplicity is exactly why RIP is a good starting point and a poor choice for large, modern networks.

CONCEPT — Hop Count and the 15-Hop Limit

RIP counts each router crossed as one hop. A destination 16 hops away is considered 'unreachable' — RIP simply refuses to believe a network can be reached. This is why RIP only fits small networks, like a single-campus lab, not a citywide ISP backbone.

◆ Periodic Updates

By default, RIP routers broadcast their entire routing table to neighbours every 30 seconds, whether or not anything changed. This is simple to implement but wastes bandwidth and slows convergence — a big reason later protocols like OSPF only send updates when something actually changes.

TRY THIS LIVE

Build a 3-router chain topology in Packet Tracer (R1-R2-R3). Enable RIP on all three, then run `debug ip rip` on R2 and watch updates arrive from both R1 and R3 every 30 seconds.

```
Router(config)# router rip
```

Enters RIP configuration mode

```
Router(config-router)# version 2
```

Switches to RIPv2, which supports subnet masks (classless routing)

```
Router(config-router)# network 192.168.1.0
```

Tells RIP to advertise this directly connected network and run RIP on matching interfaces

```
Router(config-router)# no auto-summary
```

Disables automatic summarization at classful boundaries, important in RIPv2

```
Router# show ip route rip
```

Displays only RIP-learned routes in the routing table

Feature	RIPv1	RIPv2
Subnet mask in updates	No (classful only)	Yes (classless, supports VLSM)
Update type	Broadcast	Multicast (224.0.0.9)
Authentication	Not supported	Supported (plain text or MD5)

02 Capturing and Reading RIP Updates

Watching RIP updates live is one of the best ways to truly understand routing. Every 30 seconds, you'll see a router announce its known networks — and you'll notice the hop count increasing by one each time an update crosses a router, which is exactly how RIP measures distance.

USING THIS HANDOUT FOR YOUR PRACTICE TASKS

CASE STUDY — Small Campus Networks Choosing RIP

Context — Many engineering colleges across Gujarat run modest campus networks connecting a handful of buildings — hostels, labs, and admin blocks.

Challenge — IT teams needed a routing protocol simple enough for junior staff to manage, without the complexity of area design that link-state protocols require.

What they did — For networks under 15 hops with no strict performance requirements, RIP was configured as it needs almost no design work — just enable it on each router and list the connected networks.

Result — The network self-healed when a link failed, and staff could add a new building's router by typing three lines of configuration.

Takeaway — RIP's simplicity is a genuine strength in small, non-critical networks — the limitation only shows up at scale.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — RIP configuration chapter.

DEFAULT RIP UPDATE INTERVAL

30 seconds

Source: RFC 2453 — RIP Version 2 specification.

“RIP doesn't ask which path is fastest — only which path is shortest.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Madras — Prof. Hema A Murthy — Distance vector routing module

Link — <https://www.nptel.ac.in/courses/106106091>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA lab partner. Explain why RIP has a 15-hop limit and what 'hop count = 16' actually means to the router, in simple Hinglish. Keep it under 150 words.

WHERE TO GO NEXT

You can now configure and observe RIP. Next, we move to OSPF — a protocol built for larger, more demanding networks.

Explore deeper — Read about 'split horizon' and 'route poisoning' — RIP's built-in tricks to avoid routing loops.

Try on your own — Add a fourth router to your lab and predict how the hop count will change before testing it.

Connect the dots — Session 04 introduces OSPF, which replaces hop count with a smarter, bandwidth-aware cost metric.

Going further — Research why most enterprise networks today skip RIP entirely and go straight to OSPF or EIGRP.

SESSION

04

90 MIN · ROUTING FUNDAMENTALS

Introduction to OSPF

◆ Learning Objectives

- Explain how OSPF works, including areas, LSAs, and cost
- Describe why OSPF is a link-state protocol and how that differs from RIP
- Observe OSPF adjacency formation in Packet Tracer
- Explain the role of the Designated Router (DR) and Backup Designated Router (BDR)
- List the steps required to configure OSPF in a network
- Identify why large enterprises prefer OSPF over RIP

01 OSPF — Thinking in Maps, Not Hops

OSPF (Open Shortest Path First) is a link-state protocol — instead of asking neighbours 'how many hops away is this network', every router builds its own complete map of the area and calculates the shortest path itself. It's the difference between asking for directions turn by turn versus having the entire city map open in front of you.

CONCEPT — LSAs and the Link-State Database

Routers running OSPF exchange Link-State Advertisements (LSAs) — small packets describing each router's directly connected links. Every router in the same area collects these LSAs into an identical link-state database, then runs the Dijkstra shortest-path algorithm to calculate its own best routes.

◆ Areas — Keeping the Map Manageable

In a large network, sharing one giant map with every router would overload memory and CPU. OSPF solves this by dividing the network into areas — Area 0 is always the backbone, and other areas connect through it. This is similar to how a company might organize its Ahmedabad HQ as the central hub, with each branch office as its own manageable unit.

Term	Meaning	Why It Matters
LSA	Link-State Advertisement — a router announcing its links	Builds the shared network map
Area	A group of routers sharing one link-state database	Limits map size and CPU load
Cost	OSPF's metric, based on interface bandwidth	Chooses the fastest path, not just the shortest

Term	Meaning	Why It Matters
DR / BDR	Elected routers that reduce update traffic on a segment	Prevents every router from talking to every other router

RIP (Distance-Vector)

Judges paths by hop count, shares only what neighbours tell it, slower to converge, simple to configure.

OSPF (Link-State)

Judges paths by bandwidth-based cost, builds a full network map, converges fast, needs more careful design.

TRY THIS LIVE

Open a pre-configured OSPF network in Packet Tracer. Use `show ip ospf neighbor` to watch adjacency states move from Down to Init to 2-Way to Full.

```
Router# show ip ospf neighbor
Router# show ip ospf interface
Router# show ip protocols
```

Displays neighbouring OSPF routers and their adjacency state
Shows OSPF settings and cost on each interface
Confirms OSPF is running and lists the networks it's advertising

02 Why OSPF Wins in Large Networks

CASE STUDY — Enterprise Data Centre Networks

Context — Large Indian enterprises running multiple data centres — for example, connecting an Ahmedabad primary site to a Pune or Bengaluru disaster-recovery site — need fast, reliable routing between sites.

Challenge — RIP's slow convergence and 15-hop limit made it unsuitable once the network grew past a few routers and needed near-instant failover.

What they did — Network architects deployed OSPF, dividing the network into areas aligned with each data centre, with Area 0 acting as the backbone connecting them.

Result — When a link failed, OSPF recalculated the shortest path within seconds instead of the 30-plus seconds typical of RIP, keeping applications available.

Takeaway — OSPF's link-state design trades a bit of configuration complexity for speed and scale that growing networks eventually need.

Source: Cisco Networking Academy, CCNA — OSPF Concepts and Configuration module.

OSPF ADMINISTRATIVE DISTANCE (DEFAULT)

110

Source: Cisco IOS documentation — administrative distance defaults table.

“RIP asks a neighbour for directions. OSPF draws its own map and never needs to ask again.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Kharagpur — Prof. Sujoy Ghosh — Link-state routing and OSPF module

Link — <https://nptel.ac.in/courses/106105081>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a friendly CCNA instructor. Explain OSPF areas and why Area 0 is always the backbone, using the analogy of a company's Ahmedabad head office connecting to branch offices. Keep it under 150 words, in simple Hinglish.

WHERE TO GO NEXT

You now understand how OSPF thinks. Next session, you'll configure it hands-on across a multi-area network.

Explore deeper — Look up the Dijkstra Shortest Path First algorithm — the math engine behind every OSPF route calculation.

Try on your own — Sketch your own 3-area OSPF design on paper before Session 05, then compare it to what you build in the lab.

Connect the dots — Session 05 walks through full OSPF configuration across multiple areas, plus simulating a router failure.

Going further — Research OSPFv3, which extends OSPF to support IPv6 networks.

SESSION

05

90 MIN · ROUTING FUNDAMENTALS

Configuring OSPF

◆ Learning Objectives

- Configure OSPF on a network with multiple areas
- Correctly use wildcard masks in the network command
- Verify OSPF neighbor adjacency and area assignment
- Simulate a router failure and observe OSPF reconvergence
- Troubleshoot common OSPF adjacency failures
- Explain how router-id is chosen and why it matters

01 From Concept to Configuration

In Session 04 you learned how OSPF thinks. Today you configure it. The core commands are short, but one detail trips up almost every beginner: OSPF's network command uses a wildcard mask, which is the reverse of a normal subnet mask. Get comfortable inverting subnet masks before you start typing.

CONCEPT — Wildcard Masks

A subnet mask of 255.255.255.0 becomes a wildcard mask of 0.0.0.255 in OSPF. The '0' bits mean 'must match exactly', and the '255' bits mean 'don't care'. This single reversal is the most common source of OSPF configuration mistakes for new learners.

```
Router(config)# router ospf 1
```

Starts OSPF process ID 1 — this number only has local significance on this router

```
Router(config-router)# router-id 1.1.1.1
```

Manually sets a stable router ID instead of relying on interface IPs

```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

Enables OSPF on matching interfaces and places them in Area 0 (backbone)

```
Router(config-router)# network 10.0.0.0 0.0.0.3 area 1
```

Enables OSPF on a second interface and places it in Area 1

```
Router# show ip ospf neighbor
```

Confirms adjacency has reached the 'Full' state with each neighbour

◆ The Router ID

Every OSPF router needs a unique router ID to identify itself in the network. If you don't set one manually, OSPF picks the highest IP address among loopback interfaces, or failing that, the highest IP among active interfaces. In production networks, engineers almost always set the router-id manually to avoid it changing unexpectedly after a reboot.

TRY THIS LIVE

Configure OSPF on a 3-router network spanning two areas (Area 0 and Area 1). Verify with `show ip ospf neighbor` that every expected adjacency shows state 'FULL'.

Neighbor State	What It Means
Down	No hello packets received yet
Init	Hello received, but router doesn't see itself listed back yet
2-Way	Bidirectional communication confirmed
Full	Databases fully synchronized — adjacency complete

WARNING — Common OSPF Mistakes

Mismatched hello/dead timers, mismatched area numbers on the same link, and wrong wildcard masks are the top three reasons OSPF neighbors get stuck in a state other than Full. Always check both sides of a link when troubleshooting.

02 Simulating Failure and Watching Reconvergence

The real test of any dynamic routing protocol is what happens when something breaks. Today you'll intentionally shut down a link between two routers mid-lab and watch OSPF discover the failure, remove the affected route, and recalculate a backup path — all without any manual intervention.

USING THIS HANDOUT FOR YOUR PRACTICE TASKS

CASE STUDY — Multi-Site Educational Networks

Context — Institutions running multiple campuses — for example, a training provider with centres across Ahmedabad, Surat, Rajkot, and Vadodara — often connect sites through a shared backbone.

Challenge — A single provider link failure between two centres should not disconnect their shared learning management or attendance systems.

What they did — Network teams designed each centre as its own OSPF area connected through Area 0, with redundant links between key sites.

Result — When a primary link failed during testing, OSPF recalculated a backup path within seconds, and only a brief blip was visible to end users.

Takeaway — Multi-area OSPF design turns a physical redundancy investment into an actual, automatic failover capability.

Source: Cisco Networking Academy, CCNA — OSPF Multiarea Configuration module.

TYPICAL OSPF HELLO INTERVAL ON BROADCAST NETWORKS

10 seconds

Source: Cisco IOS documentation — OSPF default timers.

“A routing protocol that only works when nothing breaks isn't worth deploying.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Demystifying Networking, IIT Bombay — Prof. Sridhar Iyer & Prof. Ashutosh Raina — Routing configuration and troubleshooting module

Link — <https://nptel.ac.in/courses/106101209>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA lab troubleshooting buddy. Explain three common reasons OSPF neighbors get stuck in the 'Init' state, in simple Hinglish, with the fix for each. Keep it under 150 words.

WHERE TO GO NEXT

You've now configured, verified, and stress-tested OSPF. Next, we shift from routing to a different layer — wireless networking.

Explore deeper — Look up 'OSPF LSA types 1 through 5' to understand what information actually gets exchanged.

Try on your own — Deliberately mismatch the area number on one router's interface and observe exactly what error message or symptom appears.

Connect the dots — Session 06 moves to wireless standards and configuration, still building on today's mindset of verify-then-trust.

Going further — Research OSPF authentication (MD5) and why production networks enable it on every OSPF-enabled interface.

SESSION

06

90 MIN · WIRELESS NETWORKING

Wireless Standards & Configuration

Learning Objectives

- Identify the major 802.11 wireless standards and their key differences
- Configure basic wireless setup and channel selection
- Explain how to avoid channel interference in dense environments
- Describe wireless security options from WEP to WPA3
- Configure MAC filtering and basic access control lists for a wireless network
- Evaluate trade-offs between wireless range, speed, and interference

01 From Wires to Waves

Everything so far in this module has assumed a cable between devices. Wireless networking removes that cable, but it doesn't remove the underlying rules — routing, addressing, and switching still apply once traffic reaches the wired network. What changes is the physical layer: radio waves instead of copper or fibre, which brings its own set of standards and problems.

CONCEPT — 802.11 Standards

IEEE 802.11 is the family of wireless standards that defines Wi-Fi. Each version — a, b, g, n, ac, ax — improves speed, range, or efficiency. A home router in an Ahmedabad apartment supporting 802.11ax (Wi-Fi 6) can serve far more devices simultaneously than an older 802.11n router, which matters a lot in dense flats and offices.

Standard	Common Name	Max Speed (approx.)	Band
802.11g	Wi-Fi 3	54 Mbps	2.4 GHz
802.11n	Wi-Fi 4	600 Mbps	2.4 / 5 GHz
802.11ac	Wi-Fi 5	3.5 Gbps	5 GHz
802.11ax	Wi-Fi 6	9.6 Gbps	2.4 / 5 / 6 GHz

Channel Selection and Interference

The 2.4 GHz band has only 3 non-overlapping channels (1, 6, and 11) in most regions. In a crowded apartment building, dozens of routers competing for the same channels causes real slowdowns — this is why 5 GHz, with far more non-overlapping channels, is preferred wherever devices support it.

TRY THIS LIVE

Use a Wi-Fi analyzer app on your phone to scan nearby networks. Note how many networks are crowded onto channels 1, 6, and 11 versus spread across 5 GHz channels.

<code>WLC(config)# wlan create WLAN-01 TOPS-LAB</code>	Creates a new wireless LAN with SSID 'TOPS-LAB'
<code>WLC(config)# ap channel 6</code>	Manually assigns channel 6 to reduce overlap with nearby networks
<code>WLC(config)# wlan security wpa wpa2 enable WLAN-01</code>	Enables WPA2 security on the wireless LAN
<code>WLC(config)# mac-filter add 00:1A:2B:3C:4D:5E</code>	Allows a specific device by its MAC address, denying all others

02 Securing the Airwaves

Unlike a wired cable, radio signals spill beyond walls — anyone within range can attempt to connect. Wireless security has evolved through several generations, each fixing weaknesses found in the last.

WEP (Legacy — Avoid)

Uses a weak, easily crackable encryption key. Considered insecure and should never be used in a new deployment.

WPA2 / WPA3 (Current Standard)

Uses strong AES encryption, with WPA3 adding protection against offline password-guessing attacks. This is the minimum standard for any modern network.

WARNING — MAC Filtering Is Not Real Security

MAC addresses can be spoofed in seconds with free software. MAC filtering is useful for basic access control on a small trusted network, but should never be relied on as the only line of defence — always pair it with WPA2/WPA3 encryption.

CASE STUDY — Campus Wi-Fi Rollouts in India

Context — Colleges and training centres across India have rapidly expanded Wi-Fi coverage to support laptops, phones, and lab devices across large buildings.

Challenge — Dense student areas like libraries and hostels overwhelmed early single-router deployments, causing dropped connections during peak hours.

What they did — IT teams redesigned coverage using multiple access points on non-overlapping channels, moved most devices to the 5 GHz band, and enforced WPA2-Enterprise authentication tied to student login credentials.

Result — Connection drops during peak hours reduced significantly, and administrators gained the ability to trace network activity back to individual student logins rather than shared passwords.

Takeaway — Wireless capacity planning — not just coverage — is what makes a campus network usable at scale.

Source: Cisco Networking Academy, CCNA — Wireless LAN Fundamentals module.

NON-OVERLAPPING CHANNELS AVAILABLE ON 2.4 GHZ

3

Source: IEEE 802.11 standard — 2.4 GHz channel allocation.

“A wireless network is only as secure as the weakest encryption standard it still allows.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks and Internet Protocol, IIT Kharagpur — Wireless LAN and 802.11 module

Link — <https://nptel.ac.in/courses/106105183>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a home networking consultant. Explain in simple Hinglish why 5 GHz Wi-Fi is often faster but has shorter range than 2.4 GHz, using an example of a typical Ahmedabad flat. Keep it under 150 words.

WHERE TO GO NEXT

You now understand wireless standards, channels, and security. Next, we return to the wired side with switching fundamentals.

Explore deeper — Look up the difference between WPA2-Personal and WPA2-Enterprise authentication.

Try on your own — Check your own home router's settings and identify which Wi-Fi standard, channel, and security type it currently uses.

Connect the dots — Session 07 begins the switching block, covering how a switch learns and forwards frames — a very different job from a router.

Going further — Research Wi-Fi 6E and Wi-Fi 7, and how the newly opened 6 GHz band changes channel availability.

SESSION

07

90 MIN · SWITCHING FUNDAMENTALS

Switching Basics

◆ Learning Objectives

- Explain how a switch operates at Layer 2 using MAC addresses
- Describe the MAC address table and how it is built
- Observe MAC table population in a simulated network
- Explain the three frame forwarding behaviours: forward, flood, and filter
- Compare switches to routers and hubs in terms of function
- Troubleshoot a switch with an unexpectedly empty or incorrect MAC table

01 A Switch's One Job: Learn and Forward

While a router moves traffic between different networks, a switch's job is narrower and faster: move frames between devices on the same network, using MAC addresses instead of IP addresses. A switch in a TOPS lab connecting 24 student laptops has to decide, many times a second, exactly which port to send each frame out of.

CONCEPT — The MAC Address Table

A switch builds a MAC address table by watching the source MAC address of every frame that enters a port. Over time it learns 'Laptop A is on Port 3, Printer B is on Port 8' — and uses that table to send future frames directly to the right port instead of everywhere.

◆ Forward, Flood, or Filter

When a frame arrives, the switch checks its MAC table for the destination address. If it's known, the switch forwards the frame out that one port only. If it's unknown, the switch floods the frame out every port except the one it arrived on. If the source and destination are on the same port, the switch filters — meaning it drops the frame, since no forwarding is needed.

Action	When It Happens	Result
Forward	Destination MAC is known in the table	Sent out one specific port
Flood	Destination MAC is unknown or is a broadcast	Sent out every port except the source port
Filter	Source and destination are on the same port	Frame is dropped — same-segment traffic doesn't need forwarding

TRY THIS LIVE

In Packet Tracer, connect 3 PCs to one switch. Before any traffic flows, check show mac address-table — it will be empty. Ping between two PCs, then check the table again and see the entries appear.

```
Switch# show mac address-table
```

Displays all learned MAC-to-port mappings

```
Switch# show mac address-table dynamic
```

Shows only dynamically learned entries (not static or system entries)

```
Switch# clear mac address-table dynamic
```

Clears learned entries, forcing the switch to relearn from scratch

```
Switch(config-if)# switchport port-security
```

Enables port security to limit which MAC addresses are allowed on a port

02 Switch vs Hub vs Router

Understanding what a switch is not helps clarify what it is. A hub, now mostly obsolete, floods every frame out every port with no learning at all. A router operates one layer up, using IP addresses to move traffic between different networks entirely. A switch sits precisely in between: smarter than a hub, but only within a single network.

Hub (Legacy)

Dumb repeater — floods every frame to every port, causing collisions and wasted bandwidth. No learning, no filtering.

Switch

Learns MAC addresses per port, forwards frames intelligently, and creates separate collision domains on every port.

CASE STUDY — Retail Store Network Upgrades

Context — Retail chains upgrading old shared-hub networks to modern switched networks in their back-office and billing systems.

Challenge — Hub-based networks caused frequent collisions and slow billing terminal response during peak shopping hours, especially with many POS devices active at once.

What they did — IT teams replaced hubs with managed switches, letting each device get its own dedicated collision-free connection with MAC-based forwarding.

Result — Billing terminal response times improved noticeably, and network troubleshooting became easier since the MAC address table could pinpoint exactly which port a problem device was connected to.

Takeaway — The jump from hub to switch is one of the highest-impact, lowest-cost upgrades a small network can make.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — Switching Concepts chapter.

COLLISION DOMAINS CREATED PER SWITCH PORT

1

Source: Cisco Networking Academy — Ethernet switching fundamentals.

“A switch doesn't guess where a frame should go — it remembers.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Kharagpur — Prof. Ajit Pal — Data link layer and switching module

Link — <https://nptel.ac.in/courses/106105080>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a friendly CCNA lab partner. Explain in simple Hinglish how a switch 'learns' MAC addresses, using the analogy of a security guard remembering which flat each visitor belongs to in an Ahmedabad apartment building. Keep it under 150 words.

WHERE TO GO NEXT

You now understand how a single switch learns and forwards. Next, we look at how switches divide one physical network into multiple logical ones — VLANs.

Explore deeper — Look up 'collision domain vs broadcast domain' and how switches affect each differently.

Try on your own — Add a fourth PC to your lab switch and observe how quickly the MAC table grows to include it.

Connect the dots — Session 08 introduces VLANs, which let a single switch behave like several separate switches.

Going further — Research Spanning Tree Protocol (STP) and why it exists to prevent loops when switches are connected redundantly.

SESSION

08

90 MIN · SWITCHING FUNDAMENTALS

VLAN Concepts

◆ Learning Objectives

- Define a VLAN and explain how it segments a physical switch into logical networks
- List the benefits of VLANs: security, performance, and organization
- Create VLANs on a switch and assign ports to them
- Explain the difference between a VLAN and a physical LAN segment
- Document VLAN configurations for handover to another engineer
- Identify real use cases where VLAN segmentation solves a business problem

01 One Switch, Many Networks

By default, every port on a switch belongs to the same broadcast domain — every device can potentially see every other device's broadcast traffic. In a TOPS lab with student PCs, a trainer's laptop, and a printer all on one switch, that's fine for 10 devices. But mix in finance systems, guest Wi-Fi, and CCTV cameras in a real office, and that flat design becomes a security and performance problem.

CONCEPT — What Is a VLAN

A VLAN (Virtual LAN) is a logical grouping of ports that behaves like its own separate switch, even though they're physically on the same hardware. Devices in VLAN 10 cannot talk directly to devices in VLAN 20 without going through a router — exactly like two physically separate switches would need one.

◆ Why Segment at All

VLANs solve three problems at once: security (finance traffic never mixes with guest Wi-Fi traffic), performance (smaller broadcast domains mean less unnecessary traffic per device), and organization (grouping devices by function — HR, Finance, Labs — regardless of which physical switch or floor they sit on).

Benefit	Example Scenario
Security isolation	Guest Wi-Fi VLAN cannot reach the finance server VLAN
Reduced broadcast traffic	A broadcast in the Sales VLAN doesn't reach IT or HR devices
Flexible grouping	All HR staff placed in one VLAN, regardless of which floor they sit on

Benefit	Example Scenario
Simplified moves/adds/changes	Moving a desk to a new port just needs a VLAN reassignment, not new cabling

TRY THIS LIVE

On a Packet Tracer switch, create VLAN 10 (Sales) and VLAN 20 (IT). Assign 2 ports to each. Ping between two PCs in the same VLAN, then try pinging across VLANs — notice it fails without a router.

<code>Switch(config)# vlan 10</code>	Creates VLAN 10 in the switch's VLAN database
<code>Switch(config-vlan)# name Sales</code>	Gives VLAN 10 a readable name
<code>Switch(config)# interface fastEthernet 0/1</code>	Selects the port to configure
<code>Switch(config-if)# switchport mode access</code>	Sets the port as an access port, carrying traffic for exactly one VLAN
<code>Switch(config-if)# switchport access vlan 10</code>	Assigns this port to VLAN 10
<code>Switch# show vlan brief</code>	Displays all VLANs and which ports belong to each

02 Documenting What You Build

A VLAN design that only exists in one engineer's head becomes a problem the day that engineer is on leave. Good practice is to maintain a simple VLAN documentation table — VLAN ID, name, purpose, and assigned ports or switches — updated every time a change is made.

USING THIS HANDOUT FOR YOUR PRACTICE TASKS

CASE STUDY — Multi-Department Office Networks

Context — Growing companies in Gujarat often start with a single flat network and later realize departments like Finance, HR, and Sales need separation as headcount grows.

Challenge — A single flat network meant any compromised device could potentially see traffic across the entire office, and broadcast storms in one department slowed down everyone.

What they did — Network administrators redesigned the office switch configuration into department-based VLANs, isolating Finance and HR traffic from general Sales and guest traffic.

Result — Broadcast traffic dropped sharply within each department's VLAN, and a later security audit confirmed guest devices could no longer reach internal file servers.

Takeaway — VLANs let a company scale network security and performance without buying

additional physical switches.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — VLANs chapter.

MAXIMUM STANDARD VLAN IDS AVAILABLE

4,094

Source: IEEE 802.1Q standard — VLAN tagging specification.

“A VLAN turns one switch into as many networks as your organization actually needs.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Kharagpur — Prof. Sujoy Ghosh — LAN switching and VLAN module

Link — <https://nptel.ac.in/courses/106105081>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA trainer. Explain why two devices in different VLANs cannot talk to each other without a router, using a simple Hinglish analogy of two separate buildings needing a connecting bridge. Keep it under 150 words.

WHERE TO GO NEXT

You can now create and assign VLANs. Next, you'll learn how to carry multiple VLANs across a single link between switches.

Explore deeper — Look up the default VLAN (VLAN 1) and why best practice is to avoid using it for real traffic.

Try on your own — Add a third VLAN to your lab and move one PC into it — observe how quickly it loses reachability to its old VLAN neighbours.

Connect the dots — Session 09 covers VLAN trunking, which lets one cable between switches carry traffic for every VLAN at once.

Going further — Research Voice VLANs and why IP phones are often placed in a separate VLAN from data devices on the same port.

SESSION

09

90 MIN · SWITCHING FUNDAMENTALS

VLAN Trunking and Access Ports

◆ Learning Objectives

- Distinguish between access ports and trunk ports
- Explain how 802.1Q trunking tags frames with VLAN information
- Configure trunk ports on a switch
- Simulate communication between VLANs using trunk ports across two switches
- Explain the purpose of the native VLAN on a trunk
- Troubleshoot a trunk link that isn't passing traffic for a specific VLAN

01 One Cable, Many VLANs

Session 08 assigned single VLANs to single ports — fine for end devices, but impractical between two switches carrying traffic for ten different VLANs. Running ten separate cables between switches, one per VLAN, doesn't scale. Trunking solves this by letting one link carry traffic for every VLAN at once.

CONCEPT — Access Ports vs Trunk Ports

An access port belongs to exactly one VLAN and connects to end devices like PCs or printers, which don't understand VLAN tags. A trunk port carries traffic for multiple VLANs simultaneously and connects switches to each other, or a switch to a router or access point that needs multi-VLAN awareness.

◆ How Trunking Actually Works — 802.1Q Tagging

The IEEE 802.1Q standard defines how a trunk port tags each frame with a small VLAN ID field before sending it across the trunk. The receiving switch reads that tag and knows exactly which VLAN the frame belongs to, then removes the tag before delivering it to an access port — end devices never see the tag at all.

Port Type	Carries	Typical Use
Access port	Traffic for one VLAN only, untagged	Connecting a PC, printer, or IP phone
Trunk port	Traffic for multiple VLANs, 802.1Q tagged	Connecting two switches, or a switch to a router

TRY THIS LIVE

Connect two switches in Packet Tracer. Configure the link between them as a trunk carrying

VLAN 10 and VLAN 20. Place one PC from each VLAN on each switch, then confirm same-VLAN PCs can ping each other across the trunk.

<code>Switch(config)# interface gigabitEthernet 0/1</code>	Selects the port connecting to the other switch
<code>Switch(config-if)# switchport mode trunk</code>	Sets the port to trunk mode, carrying multiple VLANs
<code>Switch(config-if)# switchport trunk allowed vlan 10,20</code>	
Restricts the trunk to only carry the VLANs actually needed	
<code>Switch(config-if)# switchport trunk native vlan 99</code>	
Sets a non-default native VLAN for untagged trunk traffic, a security best practice	
<code>Switch# show interfaces trunk</code>	Displays all trunk ports and which VLANs are allowed on each

02 The Native VLAN — A Common Gotcha

Every trunk has a native VLAN — traffic for this VLAN is sent untagged across the trunk. By default this is VLAN 1, but leaving it at the default is a known security risk, since an attacker could potentially inject untagged traffic. Best practice is to change the native VLAN to an unused VLAN ID and match it exactly on both ends of the trunk.

WARNING — Native VLAN Mismatch

If two switches on either end of a trunk have different native VLAN settings, VLAN traffic can leak between VLANs or trigger a 'native VLAN mismatch' warning. Always verify native VLAN settings match on both sides using `show interfaces trunk`.

CASE STUDY — Multi-Floor Office Trunking

Context — A growing company occupying multiple floors of an office building in Ahmedabad needed every floor's switch to reach every department's VLAN.

Challenge — Running separate cables per VLAN between floor switches and the core switch would have required dozens of physical links.

What they did — Network engineers configured a single trunk link between each floor switch and the core switch, carrying all department VLANs (Finance, HR, Sales, Guest) over one cable per floor.

Result — Cabling requirements dropped from many links per floor to just one trunk, while devices in each VLAN remained just as isolated as before.

Takeaway — Trunking turns VLAN segmentation from a cabling nightmare into a clean, scalable design.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — VLAN Trunking chapter.

VLAN ID FIELD SIZE IN AN 802.1Q TAG

12 bits

Source: IEEE 802.1Q standard — VLAN tag header specification.

“A trunk port doesn't erase VLAN boundaries — it just carries them all at once, correctly labelled.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks and Internet Protocol, IIT Kharagpur — VLAN trunking and 802.1Q module

Link — <https://nptel.ac.in/courses/106105183>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA lab partner. Explain 802.1Q VLAN tagging in simple Hinglish using the analogy of colour-coded courier parcels being sorted at a hub. Keep it under 150 words.

WHERE TO GO NEXT

You now know how VLANs travel between switches. Next, you'll learn how devices in different VLANs actually communicate — inter-VLAN routing.

Explore deeper — Look up 'VLAN pruning' and how it reduces unnecessary trunk traffic in large networks.

Try on your own — Add a third VLAN to your two-switch trunk lab and verify it passes correctly without disturbing the existing two.

Connect the dots — Session 10 introduces inter-VLAN routing, the missing piece that lets VLAN 10 and VLAN 20 finally talk to each other.

Going further — Research Dynamic Trunking Protocol (DTP) and why many security guidelines recommend disabling it in production.

SESSION

10

90 MIN · SWITCHING FUNDAMENTALS

Inter-VLAN Routing Basics

◆ Learning Objectives

- Explain why VLANs cannot communicate with each other without routing
- Compare Router-on-a-Stick and Layer 3 switch methods of inter-VLAN routing
- Configure router sub-interfaces for VLAN routing (Router-on-a-Stick)
- Configure Switch Virtual Interfaces (SVIs) on a Layer 3 switch
- Troubleshoot inter-VLAN routing issues in a simulated scenario
- Evaluate which inter-VLAN routing method fits a given network size

01 The Problem VLANs Create

VLANs solved isolation — but isolation is only useful when it's controlled, not absolute. A Sales VLAN and a Finance VLAN still need to reach a shared file server or the internet. Since VLANs are, by design, separate broadcast domains, something has to route between them. That something is either a router or a Layer 3 switch.

CONCEPT — Why a Router Is Needed

Switches operate at Layer 2, using MAC addresses within a single VLAN. Moving traffic between VLAN 10 and VLAN 20 means moving between two separate logical networks — a Layer 3 (routing) decision. This is true even though both VLANs might sit on the very same physical switch.

◆ Method 1 — Router-on-a-Stick

Router-on-a-Stick uses a single physical router interface, split into multiple logical sub-interfaces — one per VLAN. The link from the switch to the router is a trunk, carrying all VLANs, and the router's sub-interfaces each handle routing for their own VLAN's traffic.

```
Router(config)# interface gigabitEthernet 0/0.10
```

Creates a sub-interface for VLAN 10

```
Router(config-subif)# encapsulation dot1Q 10
```

Tags this sub-interface to handle 802.1Q traffic for VLAN 10

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

Assigns the gateway IP address for VLAN 10 devices

```
Router(config)# interface gigabitEthernet 0/0.20
```

Creates a second sub-interface for VLAN 20

```
Router(config-subif)# encapsulation dot1Q 20
```

Tags this sub-interface to handle 802.1Q traffic for VLAN 20

♦ Method 2 — Layer 3 Switch Routing (SVIs)

A Layer 3 switch skips the external router entirely. Each VLAN gets a Switch Virtual Interface (SVI) — a virtual routed interface living directly on the switch. Once IP routing is enabled, the switch itself routes between VLANs at very high speed, since it's done in hardware rather than through an external device.

<code>Switch(config)# ip routing</code>	Enables Layer 3 routing capability on the switch
<code>Switch(config)# interface vlan 10</code>	Creates the SVI for VLAN 10
<code>Switch(config-if)# ip address 192.168.10.1 255.255.255.0</code>	
Assigns the gateway IP address for VLAN 10 devices	
<code>Switch(config)# interface vlan 20</code>	Creates the SVI for VLAN 20
<code>Switch# show ip route</code>	Confirms both VLAN subnets appear as directly connected routes

Router-on-a-Stick

Uses an external router with sub-interfaces.
Lower cost if a router is already available, but limited by the single physical link's bandwidth.

Layer 3 Switch (SVIs)

Routes in hardware at very high speed, scales better for many VLANs, but requires a Layer 3-capable switch, which costs more than a basic Layer 2 switch.

02 Troubleshooting Inter-VLAN Routing

TRY THIS LIVE

Build a Router-on-a-Stick lab in Packet Tracer with two VLANs and one router. Ping between a VLAN 10 PC and a VLAN 20 PC. If it fails, check: trunk configured correctly, correct encapsulation VLAN IDs, and correct default gateway on each PC.

CASE STUDY — Growing SME Office Networks

Context — A mid-size manufacturing company in Vadodara expanded from a flat office network to a VLAN-segmented design as departments grew.

Challenge — After introducing VLANs for Finance, Production, and Admin, staff in different departments could no longer reach a shared ERP server, breaking daily operations.

What they did — IT staff enabled Layer 3 routing on the core switch and configured an SVI for each VLAN, giving every department a routed gateway back to the shared ERP server's VLAN.

Result — Cross-department access to the ERP server was restored, while VLAN isolation between departments for all other traffic remained intact.

Takeaway — VLAN segmentation and inter-VLAN routing must be planned together — isolating VLANs without a routing plan breaks legitimate business traffic.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — Inter-VLAN Routing chapter.

TYPICAL MAX SVIS ON A MID-RANGE LAYER 3 SWITCH

1,000+

Source: Cisco Catalyst switch configuration guides — SVI scalability limits.

“VLANs build the walls. Inter-VLAN routing builds the doors — on purpose, and only where you want them.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks, IIT Madras — Prof. Hema A Murthy — Layer 3 switching and inter-VLAN routing module

Link — <https://www.nptel.ac.in/courses/106106091>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA trainer. Explain in simple Hinglish the difference between Router-on-a-Stick and Layer 3 switch routing, and when a small business would pick one over the other. Keep it under 150 words.

WHERE TO GO NEXT

You've now completed the full routing-and-switching arc: static routes, dynamic protocols, wireless, VLANs, and inter-VLAN routing. Next, we consolidate everything in a weekly review.

Explore deeper — Compare the cost of a Layer 3 switch versus a router with equivalent throughput for your own reference.

Try on your own — Convert your Router-on-a-Stick lab into a Layer 3 switch lab and confirm both achieve the same ping results.

Connect the dots — Session 11 is a hands-on review combining VLANs and inter-VLAN routing into one complete network build.

Going further — Research First Hop Redundancy Protocols (HSRP, VRRP) — used to make a VLAN's gateway itself redundant.

SESSION

11

120 MIN · CONSOLIDATION & PRACTICE

Weekly Review and Practice

◆ Learning Objectives

- Combine static routing, dynamic routing, VLANs, and inter-VLAN routing into one network build
- Design and build a network with multiple VLANs and working inter-VLAN routing
- Verify end-to-end connectivity across the entire topology
- Produce a clear setup report documenting the completed network
- Self-assess understanding across all Module 1 topics before moving to WAN technologies
- Identify personal weak spots to revisit before the next module

01 Bringing It All Together

Over the last ten sessions you've built individual pieces: static routes, RIP, OSPF, wireless basics, VLANs, trunking, and inter-VLAN routing. Today's session has no new theory — it's a single combined lab where all of it has to work together, the way it would in a real small-business network.

CONCEPT — Why Integration Labs Matter

In real networks, problems rarely come from a single misunderstood topic — they come from the interactions between topics. A VLAN misconfigured on one switch can look exactly like an OSPF adjacency failure until you trace it properly. Today builds that end-to-end troubleshooting instinct.

◆ Today's Build — The Full Picture

You'll build a topology with at least two switches, one router (or Layer 3 switch), and multiple VLANs — with OSPF running between routers if more than one is used, and full inter-VLAN routing so every VLAN can reach every other permitted VLAN.

Component	What You Must Include	Session It Came From
Routing	At least one static route AND OSPF between two routers	Sessions 01, 04, 05
VLANs	Minimum 3 VLANs across 2 switches	Session 08
Trunking	802.1Q trunk carrying all VLANs between switches	Session 09
Inter-VLAN routing	Router-on-a-Stick or SVI-based routing between VLANs	Session 10

TRY THIS LIVE — 2-Hour Build

In Packet Tracer, create a network with multiple VLANs and configure inter-VLAN routing end to end. Confirm every VLAN can ping every other VLAN it's supposed to reach, and that OSPF-learned routes appear correctly with `show ip route`.

<code>Router# show ip route</code>	Confirms all expected routes — static, connected, and OSPF — are present
<code>Switch# show vlan brief</code>	Confirms every VLAN exists and has the correct ports assigned
<code>Switch# show interfaces trunk</code>	Confirms trunk links are carrying all required VLANs
<code>Router# show ip ospf neighbor</code>	Confirms OSPF adjacencies are in the Full state
<code>PC> tracert 192.168.20.10</code>	Traces the path a packet takes across VLANs and routers to reach its destination

02 Reporting and Self-Assessment

A network that works but isn't documented is only half finished from a professional standpoint. Today you'll also produce a short setup report — the kind a junior engineer would hand to a senior for review before the network goes live.

USING THIS HANDOUT FOR YOUR PRACTICE TASKS

CASE STUDY — Junior Engineer Onboarding Projects

Context — Many Indian IT services companies use a small integrated lab build like this as a final onboarding exercise for junior network engineers before assigning them live client work.

Challenge — New hires often understand individual topics in isolation but struggle when asked to design and troubleshoot a network combining routing, VLANs, and trunking together.

What they did — Training teams designed a single capstone lab requiring trainees to combine static routing, dynamic routing, VLANs, and inter-VLAN routing, then present a written setup report.

Result — Trainees who completed this kind of integration exercise were consistently faster at diagnosing real client network issues in their first few months on the job.

Takeaway — Integration practice — not just topic-by-topic learning — is what actually builds job-ready troubleshooting skill.

Source: Cisco Networking Academy, CCNA Routing and Switching Essentials — Course Summary and Skills Assessment guidance.

TOPICS COMBINED IN TODAY'S INTEGRATION LAB

4

Static routing, OSPF, VLANs, and inter-VLAN routing — the full arc of Module 1.

“You don't really know a topic until you've had to make it work alongside three others that could just as easily break it.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Advanced Computer Networks, IIT Indore & IIT Gandhinagar — Network design fundamentals module

Link — <https://nptel.ac.in/courses/106106243>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a CCNA exam coach. Ask me five quick review questions covering static routing, OSPF, VLANs, and inter-VLAN routing, one at a time, waiting for my answer before giving feedback. Keep questions in simple Hinglish.

WHERE TO GO NEXT

Module 1 (Routing and Switching) is complete. You now have the foundation every other CCNA topic builds on.

Explore deeper — Re-read your own Session 05 and Session 10 notes — convergence and inter-VLAN routing are the two topics students most often need to revisit.

Try on your own — Rebuild today's lab from scratch, without referring back to earlier sessions, to test how much you can now do from memory.

Connect the dots — Module 2 moves into WAN Technologies — MPLS, VPNs, and NAT/PAT — building directly on the routing foundation from this module.

Going further — Research the CCNA exam blueprint's weighting for Network Fundamentals vs Network Access, to see how today's topics map to real exam objectives.

REVIEW



INTERVIEW PREP · MODULE 1

Interview Questions

♦ CCNA - Routing and Switching

Rehearse these out loud and aim for answers around 60 seconds each. The hint is just a nudge — try answering before you look at it, and use the model answer afterward to check your understanding, not to memorize word-for-word.

1

What is static routing, and in what situations is it commonly used?

Think about networks with simple, unchanging topologies.

MODEL ANSWER

Static routing means manually entering routes into a router's routing table instead of letting a protocol discover them. It's commonly used in small, stable networks where the topology rarely changes, such as a stub network with a single link to the internet, or as a backup/default route in larger networks. Because it doesn't need to exchange updates, it uses no extra bandwidth or CPU.

2

List two advantages and two disadvantages of using static routes.

MODEL ANSWER

Advantages: it's predictable and secure since routes never change on their own, and it uses no CPU or bandwidth for route calculation/updates. Disadvantages: it doesn't scale well because every route must be entered by hand, and it doesn't automatically adapt if a link goes down, causing an outage until someone manually fixes it.

3

How would you configure a static route on a Cisco router using the CLI? Mention the key command syntax.

Recall the 'ip route' command and its parameters.

MODEL ANSWER

You configure a static route with the 'ip route' command in global configuration mode: `ip route <destination-network> <subnet-mask> <next-hop-ip-or-exit-interface>`. For example, `'ip route 192.168.2.0 255.255.255.0 192.168.1.2'` tells the router how to reach the 192.168.2.0/24 network via the next-hop 192.168.1.2. You can verify it afterward with 'show ip route'.

4

If you have two routers connected in Cisco Packet Tracer, what steps would you follow to enable communication between their networks using static routes?

MODEL ANSWER

First cable and IP-address the routers' interfaces so each router can reach its directly connected neighbor. Then on each router add a static route pointing to the other router's remote network, using either the next-hop IP or the exit interface. Finally verify with 'show ip route' and test with 'ping' from a PC or router interface on one side to a destination on the other side.

5

What is dynamic routing, and how does it differ from static routing?*Focus on automation and adaptability.***MODEL ANSWER**

Dynamic routing uses a routing protocol (like RIP or OSPF) so routers automatically learn about networks and update their tables when the topology changes. Static routing requires an administrator to manually add every route. Dynamic routing adapts to link failures and topology changes on its own, while static routing stays fixed until someone edits it, but dynamic routing costs more CPU/bandwidth and is more complex to secure.

6

Explain a scenario where dynamic routing would be preferred over static routing.**MODEL ANSWER**

Dynamic routing is preferred in large or frequently changing networks, for example a company with dozens of branch offices and multiple redundant links. If a WAN link fails, a dynamic protocol like OSPF automatically recalculates and reroutes traffic through a backup path within seconds, which would be impractical to do manually with static routes at that scale.

7

What does RIP stand for, and what metric does it use to determine the best path?*Think about hop count.***MODEL ANSWER**

RIP stands for Routing Information Protocol. It uses hop count as its metric, where each router a packet passes through counts as one hop, and RIP considers any path with more than 15 hops unreachable. The route with the lowest hop count is chosen as the best path.

8

Describe what is meant by convergence in RIP.**MODEL ANSWER**

Convergence in RIP is the state where all routers in the network have consistent, up-to-date routing information after a change in topology. RIP sends periodic updates (every 30 seconds by default) and uses techniques like split horizon and hold-down timers to prevent routing loops while converging. RIP convergence is relatively slow compared to protocols like OSPF because it relies on periodic rather than triggered updates.

9

How would you configure RIP on a Cisco router? List the main commands required.**MODEL ANSWER**

On each router, enter global config mode and type 'router rip' to start the RIP process, then use 'network <classful-network-address>' for each directly connected network you want RIP to advertise. Optionally use 'version 2' to enable RIPv2 for classless support and 'no auto-summary' to disable automatic summarization. Repeat this on every router participating in RIP, then verify with 'show ip protocols' and 'show ip route'.

10

In a multi-router network running RIP, how can you observe RIP updates in a simulation tool like Packet Tracer?

MODEL ANSWER

In Packet Tracer you can switch to Simulation mode and filter for RIP packets to watch the periodic update messages travel between routers in real time. You can click on each packet to inspect its contents, including the advertised networks and hop counts. You can also use 'debug ip rip' on a real or simulated router CLI to see updates being sent and received live.

11

What is an OSPF area, and why are areas used in OSPF networks?

Think about scalability and management.

MODEL ANSWER

An OSPF area is a logical grouping of routers and links that share the same link-state database, identified by an area number. Areas are used to reduce the size of the link-state database and limit the scope of LSA flooding, which improves scalability and reduces CPU/memory load as the network grows. Area 0 (the backbone) is required and all other areas must connect to it.

12

Explain the purpose of LSAs (Link State Advertisements) in OSPF.

MODEL ANSWER

LSAs (Link-State Advertisements) are the messages OSPF routers use to describe their directly connected links and networks to the rest of the area. Every router floods its LSAs to all other routers in the same area, and each router uses the collected LSAs to build an identical link-state database, from which it calculates the shortest path using the SPF (Dijkstra) algorithm.

13

How does OSPF calculate the cost of a route?

Consider bandwidth as a key factor.

MODEL ANSWER

OSPF calculates cost as a function of bandwidth, using the formula $\text{reference-bandwidth} / \text{interface-bandwidth}$ (the default reference bandwidth is 100 Mbps). Lower cost means a more preferred path, so higher-bandwidth links get lower costs. The total path cost is the sum of the costs of all outgoing interfaces along the route, and OSPF picks the path with the lowest total cost.

14

What steps would you follow to configure OSPF on a Cisco router for a single area?

Remember the 'router ospf' and 'network' commands.

MODEL ANSWER

First enable the process with 'router ospf <process-id>' in global config mode. Then advertise each connected network using 'network <network-address> <wildcard-mask> area 0' for a single-area design. Finally verify with 'show ip ospf neighbor' to confirm adjacencies and 'show ip route' to see OSPF-learned routes.

15

How can you observe OSPF adjacency formation in Packet Tracer?**MODEL ANSWER**

In Packet Tracer's Simulation mode you can filter for OSPF (hello) packets and watch the Hello and DBD/LSA exchange between routers as they move from Down to 2-Way to Full state. You can also run 'show ip ospf neighbor' on the CLI, which lists neighboring routers and their adjacency state, confirming when a Full adjacency has formed.

16

Describe the process of configuring OSPF for multiple areas in a network.**MODEL ANSWER**

For multiple areas, first plan which networks belong to which area, keeping area 0 as the backbone that every other area must connect to. On each router, enable OSPF with 'router ospf <id>' and advertise each network with 'network <address> <wildcard> area <area-id>' matching the correct area number for that interface. Area Border Routers (ABRs) sit between two areas and need network statements for both, and you verify with 'show ip ospf neighbor' and 'show ip route'.

17

If a router in an OSPF network fails, what happens to the network and how does OSPF recover?

Focus on reconvergence and route recalculation.

MODEL ANSWER

When a router in an OSPF network fails, its neighbors detect the missing Hello packets (after the dead interval expires) and remove the adjacency. Each neighbor generates a new LSA reflecting the topology change and floods it throughout the area, so all routers update their link-state database and rerun the SPF algorithm to recalculate routes. This is called reconvergence, and traffic is rerouted through any surviving alternate paths once the new shortest paths are computed.

18

What are some common wireless standards used in networking, and what frequency bands do they operate on?**MODEL ANSWER**

Common wireless standards include 802.11a (5 GHz), 802.11b/g (2.4 GHz), 802.11n (2.4/5 GHz, Wi-Fi 4), 802.11ac (5 GHz, Wi-Fi 5), and 802.11ax (2.4/5/6 GHz, Wi-Fi 6). The 2.4 GHz band has longer range but more interference and fewer non-overlapping channels, while 5 GHz offers higher speed and less interference but shorter range.

19

When setting up a wireless network, why is channel selection important and how can interference be minimized?

Consider overlapping channels and nearby networks.

MODEL ANSWER

Channel selection matters because overlapping channels from nearby access points cause interference, collisions, and reduced throughput, especially in the crowded 2.4 GHz band. Interference can be minimized by choosing non-overlapping channels (1, 6, 11 on 2.4 GHz), performing a site survey to see what channels neighboring networks use, and preferring the less congested 5 GHz band where possible.

20

How would you configure basic wireless security settings on a wireless router?

MODEL ANSWER

Basic wireless security is configured through the router's wireless settings page by choosing a security mode such as WPA2 or WPA3 (avoiding WEP and open networks), setting a strong pre-shared key, and changing the default SSID and admin password. You should also disable WPS if not needed, since it's a common attack vector, and keep the router's firmware updated.

21

What is MAC filtering, and how can it be used to control access to a wireless network?

MODEL ANSWER

MAC filtering is a security feature where the access point maintains a list of allowed (or denied) device MAC addresses, and only devices on the allow list are permitted to associate with the network. It adds a layer of access control but is not a strong security measure on its own since MAC addresses can be spoofed, so it should be combined with WPA2/WPA3 encryption.

22

Explain how a switch uses a MAC address table to forward frames.

Think about learning and forwarding processes.

MODEL ANSWER

A switch builds a MAC address table by inspecting the source MAC address of every incoming frame and associating it with the port it arrived on. When a frame arrives for a destination MAC already in the table, the switch forwards it only out that specific port; if the destination isn't in the table, the switch floods the frame out all ports except the one it came in on. This learning and forwarding process lets a switch deliver frames efficiently instead of broadcasting everything.

23

During a simulation, how can you observe the population of the MAC address table on a switch?

MODEL ANSWER

In a simulated network you can generate traffic between two PCs (e.g., a ping) and then check the switch's MAC table using 'show mac address-table', which will show newly learned MAC-to-port mappings after the traffic passes through. Repeating this from different devices lets you watch the table populate dynamically as more hosts communicate through the switch.

24

What is a VLAN, and what are its main benefits in a switched network?*Consider segmentation and security.***MODEL ANSWER**

A VLAN (Virtual LAN) is a logical grouping of switch ports into separate broadcast domains, even if the ports are on the same physical switch. Its main benefits are segmentation (isolating traffic between departments or functions), improved security (limiting broadcast traffic and access between groups), and easier network management, since VLANs can be reorganized without rewiring.

25

How would you create a VLAN on a Cisco switch and assign specific ports to it?**MODEL ANSWER**

First create the VLAN with 'vlan <number>' followed by 'name <vlan-name>' in global config mode. Then go into the relevant interface(s) with 'interface <type/number>' (or 'interface range' for multiple ports), set the port to access mode with 'switchport mode access', and assign it to the VLAN with 'switchport access vlan <number>'.

26

After configuring VLANs, what command would you use to verify the VLAN assignment of switch ports?**MODEL ANSWER**

You verify VLAN port assignment with 'show vlan brief', which lists each VLAN along with the switch ports assigned to it. You can also check a specific interface's VLAN membership with 'show interfaces <interface> switchport', which shows the access VLAN and mode.

27

What is the difference between a trunk port and an access port on a switch?*Think about how each handles VLAN traffic.***MODEL ANSWER**

An access port carries traffic for only one VLAN and is used to connect end devices like PCs, while a trunk port carries traffic for multiple VLANs simultaneously (tagging each frame with a VLAN ID) and is used to connect switches to each other or to a router. Access ports strip VLAN tags before delivering frames to the connected device, while trunk ports preserve the tags.

28

Name two trunking protocols and explain how 802.1Q works.**MODEL ANSWER**

Two common trunking protocols are 802.1Q (an open IEEE standard) and ISL (a legacy Cisco-proprietary protocol). 802.1Q works by inserting a 4-byte VLAN tag into the Ethernet frame header that identifies which VLAN the frame belongs to, except for the native VLAN, which is sent untagged; the receiving switch reads this tag to know how to forward and process the frame.

29

How would you configure a trunk port on a Cisco switch, and why is trunking needed in a multi-VLAN environment?

MODEL ANSWER

Configure a trunk port with 'interface <type/number>', then 'switchport trunk encapsulation dot1q' (on switches that support multiple encapsulations) followed by 'switchport mode trunk'. Trunking is needed in a multi-VLAN environment because a single physical link between switches must carry traffic for many VLANs simultaneously, and tagging lets the receiving switch distinguish which VLAN each frame belongs to.

30

Describe a scenario where two devices in different VLANs need to communicate. What configuration is required to enable this communication?

Think about inter-VLAN routing.

MODEL ANSWER

Since VLANs are separate broadcast domains, devices in different VLANs cannot communicate directly at Layer 2 even if physically connected to the same switch. To enable communication you need Layer 3 routing between the VLANs, either through a router configured with sub-interfaces for each VLAN (router-on-a-stick) or a Layer 3 switch with an SVI created for each VLAN and IP routing enabled.

31

What is the 'router-on-a-stick' method for inter-VLAN routing, and how does it work?

MODEL ANSWER

Router-on-a-stick is a method where a single physical router interface is divided into multiple logical sub-interfaces, one per VLAN, connected to a switch trunk port. Each sub-interface is assigned an IP address in its VLAN's subnet and configured with 802.1Q encapsulation matching the VLAN ID, so the router can route traffic between VLANs even though only one physical link is used.

32

List the steps to configure router-on-a-stick for inter-VLAN communication on a Cisco router.

MODEL ANSWER

First connect the router's interface to a trunk port on the switch. On the router, create a sub-interface for each VLAN using 'interface <physical-interface>.<vlan-id>', set encapsulation with 'encapsulation dot1q <vlan-id>', and assign an IP address with 'ip address <address> <mask>' that matches that VLAN's subnet. Repeat for each VLAN, then verify with 'show ip interface brief' and test connectivity between hosts in different VLANs.

33

If inter-VLAN routing is not working in your network, what troubleshooting steps would you take to diagnose the issue?

MODEL ANSWER

First confirm the switch port to the router is set to trunk mode and is carrying the correct VLANs (check 'show interfaces trunk'). Then verify each router sub-interface has the correct VLAN ID, matching encapsulation, and correct IP address ('show ip interface brief' and 'show running-config'). Finally check that end devices have the correct default gateway set to their VLAN's sub-interface IP, and test with ping/traceroute at each hop.

34

How would you document your VLAN and inter-VLAN routing configuration for a network setup report?

MODEL ANSWER

A good configuration report would list each VLAN with its ID, name, and assigned switch ports, the trunk links between switches/routers along with the encapsulation used, the router or Layer 3 switch interfaces/SVIs with their IP addresses, and any verification command output (like 'show vlan brief' and 'show ip route') as evidence the setup works. Including a simple network diagram makes it much easier for others to follow.

35

Suppose you have a network with three VLANs and need to enable communication between all of them. Describe the overall process and key components involved.

MODEL ANSWER

You would create the three VLANs on the switch(es) and assign the appropriate access ports to each. You'd configure a trunk link from the switch to the router (or use a Layer 3 switch), then set up either three router sub-interfaces (router-on-a-stick) or three SVIs (Layer 3 switch), each with an IP address in its VLAN's subnet and IP routing enabled. Finally you'd verify connectivity within each VLAN and between VLANs, and confirm end devices have the correct default gateways.

36

How would you use AI tools like ChatGPT to help you write or troubleshoot Cisco router and switch configurations, and what are some reasons you should still manually check the configuration before applying it?

Think about syntax accuracy and network-specific requirements.

MODEL ANSWER

AI tools like ChatGPT can quickly generate a starting point for router/switch configuration syntax or help explain an error message, which speeds up learning and troubleshooting. However, you should always manually verify the suggested commands against your specific device's IOS version, current running-config, and actual network addressing, since AI can produce plausible-looking but incorrect or outdated syntax and doesn't know your network's specific context.

37

Imagine you configured a network with static routes, RIP, and OSPF. How would you decide which routing protocol or method to use in different parts of a larger enterprise network?

Consider network size, scalability, and administrative overhead.

MODEL ANSWER

For a larger enterprise network, static routes work well for small stub sites with only one path in/out, or as a default/backup route. RIP is largely outdated and better replaced by faster-converging protocols; OSPF is a good choice for the core and distribution layers of a mid-to-large network because it scales well with areas, converges quickly, and supports load balancing. The decision comes down to network size, how often the topology changes, and how much administrative overhead the team can support.

38

How could an AI-based network assistant help you identify and resolve VLAN configuration errors in a simulated environment, and what limitations might such tools have?

MODEL ANSWER

An AI-based network assistant could help by suggesting likely causes for common VLAN symptoms (like mismatched trunk encapsulation, wrong VLAN ID, or a port left in the wrong mode) and by explaining relevant 'show' command output. Its limitations are that it can't directly see your live device configuration or topology, so its suggestions are generic until you provide the specific configs, and it can still make mistakes, so you must validate its advice against the actual device output.

39

Explain how switch MAC tables and VLAN configurations work together to control traffic flow in a multi-VLAN network.

MODEL ANSWER

The switch's MAC address table lets it forward frames to the correct port within a VLAN by mapping learned MAC addresses to the port they were seen on, but this table is scoped per VLAN, so a switch will only forward within the same VLAN's broadcast domain. VLAN configuration determines which broadcast domain each port belongs to, meaning traffic is contained to devices in the same VLAN unless it's explicitly routed between VLANs by a router or Layer 3 switch.

40

Describe the process and commands needed to set up a wireless network with MAC filtering and assign specific VLANs to different SSIDs on a managed switch.

MODEL ANSWER

First configure the VLANs on the managed switch and assign the appropriate ports/SSIDs to each VLAN using trunk links between the switch and the wireless access point (or controller). On the access point, map each SSID to its corresponding VLAN ID so wireless clients on that SSID land in the right VLAN. Then configure MAC filtering on the access point by entering the allowed MAC addresses under the wireless security/access control settings, and verify by connecting test devices to each SSID and confirming both VLAN placement and MAC-based access control work as expected.